The Validity and Reliability of the SINBAD Classification System for Diabetic Foot Ulcers

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ABSTRACT

Diabetic foot ulcers (DFUs) are a serious and costly complication of diabetes mellitus with a global prevalence of 6.3% and cost of £8,800 per unhealed DFU in the National Health Service. The three main types of DFU are neuropathic, ischemic, and neuroischemic, with an estimated prevalence of 35%, 15%, and 50%, respectively. Because 85% of lower-limb amputations in patients with diabetes are preceded by a DFU, the task of reducing the current and future burden of DFUs on an international level is of crucial importance.

Classification of a DFU is an important and complex process with many independent variables that influence the wound severity. Correct classification of a DFU is important to prevent deterioration in the short term and lower-limb amputation in the long term. Both the accuracy of the clinician's interpretation of categorical data from a classification model and grasp of contextual risk factors can refine diagnoses.

The term SINBAD is an acronym for six independent variables: site, ischemia, neuropathy, bacterial infection, area, and depth. This system uses comprehensive parameters with strict criteria to facilitate quick and accurate clinical decisions to prevent lower-limb amputation. In addition to providing quantitative measurement, SINBAD also spotlights the multifaceted characteristics of DFUs. By evaluating the validity and reliability of the SINBAD classification system, its applicability for the assessment of DFUs and prevention of lower-limb amputation can be better understood.

KEYWORDS: DFU, diabetic foot ulcers, reliability, SINBAD, specificity, ulcers, validity

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INTRODUCTION

Diabetic foot ulcers (DFUs) are a serious and costly complication of diabetes mellitus with a global prevalence of 6.3%, annual cost of $1.5 billion in the US,[1,2] and cost of £8,800 per unhealed DFU in the UK according to the National Health Service (NHS).[3] The three main types of DFU are neuropathic, ischemic, and neuroischemic, with estimated prevalences of 35%, 15%, and 50%, respectively.[4] A systematic review in 2017 suggested that 85% of lower-limb amputations in patients with diabetes are preceded by a DFU.[1] These statistics reveal the crucial task of reducing the current and future burden of DFUs on an international level.[5,6]

Classification of a DFU is a complex process with many independent variables influencing wound severity.[7] One DFU classification tool used by clinicians in various global healthcare systems is SINBAD, which addresses six important independent variables: site, ischemia, neuropathy, bacterial infection, area, and depth.[7] The efficacy of the SINBAD classification system is dependent on the accuracy of the clinician's diagnoses.[8] In this article, the SINBAD classification system is discussed and compared with established, evidence-based alternatives for DFU assessment such as the Texas classification tool.[7] The Wagner Scale, WIfI (wound condition, ischemia, and foot infection), and, more recently, PEDIS (perfusion, extent, depth, infection, and sensation) and TIME (tissue, infection, moisture, edge) are also recognized classification tools for wound assessment; however, not all of these are related to the classification of DFUs, specifically.[9–11]

Clinimetrics is the measurement or description of clinical phenomena using indexes.[12,13] The characteristics and wider recognition of the indexes can indicate the degree of validity.[13] As a clinimetric tool, SINBAD can be used in a clinical setting by a trained healthcare professional to classify the type of ulceration present. Several studies have supported the use of SINBAD worldwide because of its notable usability in different communities. It provides a versatile framework to accelerate the

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process of ulcer detection and diagnosis and prevent further deterioration.[14] This may be particularly useful in the initial stages of assessment because with early DFU detection, clinicians can prevent the proliferation of bacteria within the wound bed.[14]

By adopting strict criteria, SINBAD can facilitate accurate and efficient clinical decisions, such as prompt referrals to a multidisciplinary foot team within an acute setting to ultimately prevent lower-limb amputation.[15] By evaluating the validity and reliability of the SINBAD classification system, its applicability in the assessment of DFUs and prevention of lower-limb amputation can be better understood.[16,17]

VALIDITY

In the context of DFU classification, validity is based on the sensitivity and specificity of the tool in addressing the clinical presentation of a DFU (Table 1).[18] Sensitivity is defined as the accuracy of the measure in detecting individuals who do not have the disease in a certain population.[18] In contrast, specificity is the ability of the tool to accurately detect the presence of a condition in a certain population.[21] Sensitivity and specificity can often have an inverse relationship: as the sensitivity of a tool increases, the specificity decreases.[18]

Face Validity

To determine clinical credibility, experts in the field of DFU management compare the face value of SINBAD against the construct.[22] For example, expert consensus from the International Working Group for the Diabetic Foot provides high face validity for SINBAD as a DFU classification tool.[23] Thus, the broad usability of the SINBAD construct in clinical practice has prompted recognition from experts in the field of DFU management. A feature of the SINBAD classification system that gives it face validity is the inclusion of both neuropathy and ischemia, providing clinicians with a wider differential diagnosis for DFUs.[8] Although SINBAD has not been used in randomized controlled trials to establish face validity, its wide endorsement by relevant national and international governing bodies and use in large cohort studies both nationally and internationally give SINBAD credibility.[7,8]

Table 1. DEFINITIONS OF VALIDITY

Measure	Definition
Validity	The degree to which a tool accurately measures what it is intended to measure[17,18]
Face validity	The measure of whether a clinimetric tool appears to have internal validity as determined by experts in a particular field[19]
Construct validity	The extent to which the clinical measure accurately assesses what it is meant to assess[14]
Content validity	The extent to which a measure represents all facets of its own construct[20]
Criterion validity	The comparison of an established measurement (eg, SINBAD) to a representation of the construct[14]

Abbreviation: SINBAD, site, ischemia, neuropathy, bacterial infection, area, and depth.

Construct Validity

The SINBAD system uses a binary score-based system for each independent variable to produce an overall score that can help determine the classification of a DFU. Leese et al[7] suggested that the quantitative binary data gathered from SINBAD is useful for clinicians because scores can easily be compared.[7] The SINBAD system was developed from the previous SAD (site, area, depth) classification system, and no statistical tests were used in its development.[24] Further, the literature suggests that no direct patient input has been used to improve the sensitivity and specificity of the SINBAD classification tool for DFUs.[7,8] Unlike the Texas classification tool, the use of both neuropathy and ischemia as categories in SINBAD contributed to the objective of improving the overall construct validity of a classification tool for DFUs and standardizing "best practice."[25]

Convergent validity. As a subtype of construct validity, convergent validity concerns whether two comparable constructs that contain similar variables correspond with each other.[26] Recent research has suggested that both SINBAD and the Texas classification system showed a "stepwise decrease" in the number of DFUs healed after an increase in the overall scores from both classification tools.[7] This occurred alongside comparable c-statistics of 0.71 and 0.72, respectively.[7] Both c-statistics being greater than 0.7 indicates good construct validity because both SINBAD and the Texas system show suitability for binary outcomes in a logistical regression model.[7] However, Leese et al[7] also found that SINBAD had a greater construct validity than the Texas tool because the specificity was higher and therefore more suitable for audit purposes.

Content Validity

In a 2017 study evaluating the impact of plantar sheer stress on DFUs, Yavuz et al[27] noted that DFU location is an important factor impacting the extent to which weight bearing can deteriorate the wound bed and borders.[27] The most common cause of foot ulceration is repetitive mechanical pressures on neuropathic plantar tissue.[28,29] Although neuropathic ulcers usually occur on the plantar metatarsal phalangeal joint areas and plantar aspect of the hallux, evidence suggests that the hindfoot is at particularly high risk for severe cases of pressure-related DFU occurrence.[30] This justifies a score of 1 for the midfoot-hindfoot areas and a score of 0 for the forefoot in the SINBAD classification tool. Higher specificity could be achieved by distinguishing the midfoot from the hindfoot because there is a lack of evidence to suggest that the midfoot is a high-risk area for the most severe cases of DFU occurrence.[30]

Tests for ischemia are important in determining the arterial supply to the foot. The most frequently used tests for pedal blood flow and detection of lower-limb ischemia are manual palpation and handheld Doppler ultrasound.[31] Obtaining an ankle-brachial pressure index is the criterion standard but is not as commonly used to determine ischemia.[32] However, in contradiction with SINBAD's high face validity, the term "reduced pedal flow" is obscure as a measurement of ischemia and open to misinterpretation by clinicians with various levels of competency.[33] One solution to this issue would be to create a clear, embedded definition of "reduced pedal flow" and accompanying score to improve test accuracy. An alternative solution, however, could be to include the combination of intermittent claudication and rest pain as key pathologic determinants of ischemia.[34] This could fill the gap of crucial missing data on the extent of lower-limb ischemia in patients with DFUs.[7] However, previous research suggests clinicians can have difficulty in determining intermittent claudication, which could increase the likelihood of inaccurate test data.[34]

Neuropathy is determined by a monofilament score of 7 or lower.[35] As with ischemia, a limitation of the SINBAD system for scoring neuropathy is a lack of specificity pertaining to the broad interpretability of the term "intact."[29] A clear numerical reflection of the words "protective sensation" and "intact," based on a monofilament score that clearly defines the starting point of neuropathy, could provide a quantitative solution to this issue.[1] Limitations of this solution are the interobserver reliability due to the variable quality of monofilament assessment technique from the clinician and subjective interpretation of the term "protective sensation."[36]

Following measurement for neuropathy, the detection of bacterial infection in DFUs can lead to preventive strategies in cases of infection-related complications such as osteomyelitis, amputation, and systemic infection.[37] According to a recent international multicenter study, understanding the extent of bacterial infection contributed to the "strongest predictors of healing" across each center ($P < .001$).[7]

In the SINBAD classification system, a DFU area measurement greater than 1 cm^2 receives a score of 1, and a DFU measuring 1 cm^2 or smaller scores 0. However, because the wound edges for measuring the diameter of the DFU are not specified within the category, the content validity is poor; the results would likely be more consistent when obtained exclusively from trained clinicians.[38] A possible solution to this limitation would involve contact measurement of the diameter of wound margins (excluding periwound tissue) using acetate tracing or the ruler from a sterile scalpel handle.[39]

Similarly, depth can also present with limitations to the content validity of SINBAD.[7] Distinguishing between the fine margins of a binary score could be a limitation when differentiating between a DFU that is reaching subcutaneous tissue versus muscle.[40] Although a probe-to-bone test is useful in determining depth, this is not mentioned in depth in the SINBAD system. Further, SINBAD does not specifically address tendons and muscles as independent variables.[8] The absence of a quantitative measurement score could suggest a degree of methodologic inconsistency.[20]

Criterion Validity

The use of systematic binary scoring in the SINBAD classification system, contrasted with the descriptive categories and grading measurements of the Texas tool, makes SINBAD structurally more effective for attaining a fast assessment and DFU diagnosis within a time-pressured clinical setting such as a hospital foot protection clinic.[24]

Concurrent validity. Concurrent validity is established by comparing the assessed clinimetric tool against the criterion standard.[7] Currently, there is no confirmed criterion standard tool with established high validity, sensitivity, and specificity for DFU classification in research.[8] However, a recent observational analysis of ulcer outcomes compared the SINBAD and Texas scoring systems for predicting ulcer outcome in a routine diabetic foot clinic.[7] Longitudinal data from 1,065 outpatients with DFUs referred between 2006 and 2018 revealed missing data for 14% of DFU size calculations using the SINBAD tool, which was attributed to the time limits of a busy routine clinic.[7] Another limitation to this study was that patients with DFUs managed in the community were not considered as part of the research design; thus, the study lacked a reflection of the big picture.[7] The ability to attain robust, quality data, even in established healthcare systems such as the NHS, may be limited.

Predictive validity. Another subtype of criterion validity, predictive validity is the ability of a test to accurately predict future performance outcomes.[6] Predictive factors in patients with DFUs are often multifactorial. As a result, the SINBAD system may demonstrate higher predictive validity than the Texas tool because SINBAD integrates six different elements within the classification system.[6] However, a previous multicenter study of 449 patients with DFUs referred to the limitations of a "variable duration of follow-up" in different international centers.[8] This is supported by several studies that suggest the ability of a classification system (eg, SINBAD) to predict future performance outcomes is dependent on external factors and internal validity.[41–43] Alsabek and Abdul Aziz[44] expanded on this to suggest that resource-poor environments can delay access to immediate DFU assessment and classification, therefore increasing the likelihood of lower-limb amputation.

RELIABILITY

If, under similar conditions, a tool can produce consistently repeatable results, then it can be considered reliable.[45] For a DFU classification system to be useful to clinical practice, it must have strong interrater and intrarater reliability.[46] Reliability ensures robust data quality and completeness. However, there is a lack of research evaluating the interrater and intrarater reliability of DFU classification tools.[47] The more accessible the usability, the more likely it is that the classification system will provide reliable quantitative data capable of intrarater correlation.[47] Further research is needed to establish the reliability of the SINBAD classification system.[7]

RESPONSIVENESS

For an evaluative classification tool such as SINBAD to be responsive, it must be able to accurately measure change over a set time period. This occurs numerically in the SINBAD tool using an overall scale (0-6) as a measure of treatment effectiveness and DFU healing progress.[7] However, the interpretability of each independent variable can increase the likelihood of errors in interrater reliability outcome data.

ACCEPTABILITY

The SINBAD classification system provides an agile, convenient, and comprehensive template of evaluation for the assessment and diagnosis of DFUs.[8] Because the assessment is based on numerical values, it provides a quantitative reflection of a DFU with no equipment required; SINBAD can be used universally and at speed.[7]

Approval of the SINBAD classification system as a useful tool in podiatry DFU care in the National Institute for Clinical Excellence's guidelines for the diabetic foot demonstrates general acceptability.[48] In addition, the National Diabetic Foot Audit included SINBAD as an acceptable tool for collating large data sets in the NHS.[25] Thus, both of these reputable governing bodies have endorsed SINBAD as a standardized system.[25,48] Internationally, the Australian guidelines on wound classification of diabetes-related foot ulcers recommend SINBAD as a "minimum standard" for audit purposes.[22] Overall, SINBAD currently has a high acceptability in clinical practice both nationally and internationally.[24]

CONCLUSIONS

The SINBAD system is the most recent DFU classification tool to gain international consensus regarding its detection of ischemic and neuropathic DFUs.[7] As an evaluative and predictive measure, SINBAD is widely accepted in the literature and by relevant governing bodies as providing a flexible and responsive strategy for the classification of DFUs, while balancing limitations of various levels of advancement in different international healthcare systems.[23,25,48]

The effective use of the SINBAD classification system is dependent on robust healthcare systems, with its usability limited to trained healthcare professionals. This is supported by the Australian guidelines on wound classification of DFUs, which states that SINBAD should be used as a "minimum standard" rather than as a criterion standard.[22] However, the good overall validity and reliability of SINBAD are important clinical assets for the goal of reducing amputation rates internationally. ●

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